Input Signals (100 Hz)

- Nominal Control Action:
  u_nom (VLA / Flow Policy)
- Manipulator Kinematics:
  Joint State (q_k, dq_k in R^16)
- Uncertainty Calibration:
  mu_cal (from ACI module)
- Object Load Invariants:
  Mass m_o, Centroid c_o

Contact Sensitivity & Grasp

- Linearized Contact Dynamics:
  Lambda_i = (0.5 dt^2 K_c
  + dt D_c) J_i
- Forward Force Mapping:
  f_{c,i}(k+1) = Lambda_i u
  + f_{c,i}^bias
- Grasp Matrix: G in R^{6 x 12}
- Nullspace Decoupling:
  G * Delta f_internal = 0

Predictive Safety Filter QP (OSQP ADMM Core)

Convex Objective Function:

$$\min_{\mathbf{u},\boldsymbol{\delta}_s,\boldsymbol{\delta}_w} \ \tfrac{1}{2}\|\mathbf{u}-\mathbf{u}_{\text{nom}}\|_{\mathbf{W}}^2 + \tfrac{1}{2}\rho_s\|\boldsymbol{\delta}_s\|_2^2 + \tfrac{1}{2}\rho_w\|\boldsymbol{\delta}_w\|_2^2$$

Weights: W = I_16, rho_s = 10^5 (Hard Barrier Penalty), rho_w = 10.0 (Wrench Compliance)

Discrete-Time Barrier & Physical Constraints:

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[1] 8-Sided Polyhedral Anti-Slip Barrier (P_8):
    A_cone,i Lambda_i u - delta_s <= (1 - gamma_slip) p_curr,i - A_cone,i f_bias,i
    (Radial Conservatism <= 7.61%, Area Conservatism <= 9.97%)

[2] Fragile Object Anti-Crush Force Ceiling:
    n_i^T Lambda_i u <= F_max - n_i^T f_bias,i       (F_max = 18.0 N)

[3] Contact Maintenance Pre-Load Threshold:
    -n_i^T Lambda_i u <= -F_min + n_i^T f_bias,i      (F_min = 1.0 N)

[4] Spatial Object Wrench Equilibrium:
    G Lambda_stack u - delta_w = w_req - G f_bias_stack

[5] Actuator Motor Acceleration Saturation:
    u_min <= u <= u_max   (|u_j| <= 30.0 rad/s^2)
```

Real-Time Decision Gate & Safety Supervisor

NOMINAL_SAFE	Solve time <= 10 ms & ||delta_s||_inf <= 1e-2 N -> Certified Forward Invariance; dispatch u*
EMERGENCY_RELAXED	||delta_s||_inf > 1e-2 N (Soft Constraint Slack Active) -> Time-bounded emergency alert; trip if steps > 5
TIMEOUT	Execution latency > 10.0 ms Watchdog Deadline -> Abort QP solve immediately; enter passive damping
CRITICAL_FAIL	Persisted relaxation (> 5 steps) or numerical fault -> Dispatch Passive Damping u_safe = -K_d * dq_k (< 50 us)

Dispatched Control Command u_k* (100 Hz Control Cycle)

Low-Level Computed Torque: tau = M(q) u* + C(q, dq) dq + g(q)